Quantum Field Theory of the σ-Field

Propagators, UV Regularisation, and the Double Copy

from Quantum Gravitational Dynamics

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Abstract

We develop the quantum field theory of the QGD σ-field. The Pais–Uhlenbeck propagator $D(k^2) = 1/k^2 - 1/(k^2 + m_Q^2)$ with $m_Q = M_{\rm Pl}/\sqrt{\kappa}$ naturally regularises UV divergences: one-loop gravitational integrals become at most logarithmically divergent, compared to the quadratic divergence in perturbative GR. We show that tree-level σ-exchange reproduces Newton's law, identify the metric $h_{\mu\nu} = -\sigma_\mu\sigma_\nu$ as the field-level realisation of the BCJ double copy, derive the gravitational Rutherford cross section, compute the σ-field vacuum energy and its relation to dark energy, and show that σ-entanglement at the horizon gives the correct area scaling for black hole entropy.

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1 The σ-Field Propagator

1.1 Derivation from the QGD action

The σ-field equation from the QGD effective action is:

$$\Box_g \sigma_\mu - \kappa \ell_Q^2 \Box_g^2 \sigma_\mu = \frac{4\pi G}{c^2} J_\mu \tag{1}$$

where $\ell_Q = \sqrt{G\hbar^2/c^4} \approx 9.6 \times 10^{-57}$ m and $\kappa \approx 2$.

In momentum space (flat background):

$$(-k^2 - \kappa \ell_Q^2 k^4)\tilde{\sigma}_\mu = \tilde{J}_\mu \tag{2}$$

The propagator is:

$$D(k^2) = \frac{-1}{k^2 + \kappa \ell_Q^2 k^4} = \frac{-1}{k^2(1 + \kappa \ell_Q^2 k^2)} \tag{3}$$

1.2 Pais–Uhlenbeck factorisation

Partial fractions give:

$$\boxed{D(k^2) = \frac{1}{k^2} - \frac{1}{k^2 + m_Q^2}} \tag{4}$$

where:

$$m_Q^2 = \frac{1}{\kappa \ell_Q^2} = \frac{c^4}{\kappa G \hbar^2} \quad \Longrightarrow \quad m_Q = \frac{M_{\rm Pl}}{\sqrt{\kappa}} \approx 0.71\, M_{\rm Pl} \approx 1.54 \times 10^{-8}\ \text{kg} \tag{5}$$

Two propagating modes:

- **Mode 1 (massless):** pole at $k^2 = 0$ with residue $+1$. This is the classical graviton, mediating long-range $1/r$ gravity.
- **Mode 2 (Planck-mass):** pole at $k^2 = -m_Q^2$ with residue -1. This is a Planck-mass excitation that decays as $e^{-m_Q r}$ with lifetime $\tau_Q \sim \hbar/(m_Q c^2) \sim 10^{-43}$ s.

The negative residue of the massive mode is the hallmark of a Pais–Uhlenbeck theory. It does *not* indicate a ghost instability: the Hamiltonian is positive-definite when quantised with the Dirac–Pauli inner product (proven in Chapter 3 via the canonical analysis of the fourth-order system).

1.3 Behaviour across scales

At $k \ll k_{\rm Pl}$: $D \approx 1/k^2$ (standard GR). At $k \sim k_{\rm Pl}$: the massive mode cancels the massless mode, and $D \to 1/(k^2 m_Q^2) \propto 1/k^4$. The propagator softens from $1/k^2$ to $1/k^4$ at the Planck scale — this is the mechanism behind UV regularisation.

Table 1: σ-field propagator at various momentum scales

$k/k_{\rm Pl}$	$D_{\rm massless}$	$D_{\rm total}$	$D_{\rm total}/D_{\rm massless}$
10^{-30}	$1/k^2$	$1/k^2$	1.000000
10^{-10}	$1/k^2$	$1/k^2$	1.000000
10^{-2}	$1/k^2$	$0.9998/k^2$	0.99980
10^{-1}	$1/k^2$	$0.980/k^2$	0.980
1	$1/k^2$	$0.333/k^2$	0.333
10	$1/k^2$	$0.005/k^2$	0.005

2 Tree-Level Scattering: Newton's Law from σ-Exchange

2.1 The gravitational vertex

The metric perturbation $h_{\mu\nu} = -\sigma_\mu\sigma_\nu$ couples to matter through $\mathcal{L}_{\rm int} = -\frac{1}{2}h_{\mu\nu}T^{\mu\nu} = \frac{1}{2}\sigma_\mu\sigma_\nu T^{\mu\nu}$.

For static point masses $T^{00}_a = M_a c^2\,\delta^{(3)}(\mathbf{x}-\mathbf{x}_a)$, the coupling is $\frac{1}{2}\sigma_0^2 M_a c^2$. The tree-level amplitude for σ_0 exchange between two static masses:

$$i\mathcal{M} = \frac{(iM_1c^2)(iM_2c^2)}{c^4/(16\pi G)} \times \frac{1}{|\mathbf{q}|^2} \tag{6}$$

where $\mathbf{q}$ is the three-momentum transfer and $c^4/(16\pi G)$ is the normalisation from the Einstein–Hilbert action.

2.2 Newton's potential

Born approximation:

$$V(r) = -\int \frac{\mathrm{d}^3q}{(2\pi)^3}\frac{16\pi G M_1 M_2}{|\mathbf{q}|^2}e^{i\mathbf{q}\cdot\mathbf{r}} = -\frac{GM_1M_2}{r} \tag{7}$$

Newton's law is recovered exactly from tree-level σ-exchange. ✓

2.3 Yukawa correction from the massive mode

Including both propagator poles:

$$V_{\rm QGD}(r) = -\frac{GM_1M_2}{r}\left(1-e^{-m_Q r}\right) \approx -\frac{GM_1M_2}{r}\left(1-e^{-r/\ell_{\rm Pl}}\right) \tag{8}$$

At $r \gg \ell_{\rm Pl}$: $V \to -GM_1M_2/r$ (Newton). At $r \sim \ell_{\rm Pl}$: the Yukawa factor suppresses the potential, regularising the singularity.

2.4 Gravitational Rutherford scattering

The differential cross section for non-relativistic gravitational scattering:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{(Gm_1m_2)^2}{4E^2\sin^4(\theta/2)} \tag{9}$$

This is the Rutherford formula with G replacing $k_e^2/(m_1m_2)$. For proton–proton scattering at $v = 1$ km/s: $\sigma_{\rm grav} \approx 3.9\times10^{-71}$ m^2 — a factor of 10^{47} weaker than Coulomb scattering.

3 UV Regularisation: One-Loop Behaviour

3.1 The standard GR divergence

The one-loop graviton self-energy in perturbative GR:

$$I_{\mathrm{GR}} = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{k^2} \frac{1}{(k-p)^2} \sim \Lambda^2 \qquad \text{(quadratic divergence)} \tag{10}$$

This is why perturbative quantum gravity is non-renormalisable: each loop introduces new divergences requiring new counterterms.

3.2 The QGD-regulated integral

In QGD, each propagator $1/k^2$ is replaced by $D(k^2) = 1/k^2 - 1/(k^2 + m_Q^2)$. The one-loop integral:

$$I_{\mathrm{QGD}} = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \left[\frac{1}{k^2} - \frac{1}{k^2 + m_Q^2} \right] \left[\frac{1}{(k-p)^2} - \frac{1}{(k-p)^2 + m_Q^2} \right] \tag{11}$$

Expanding:

$$I_{\mathrm{QGD}} = I_{00} - I_{0m} - I_{m0} + I_{mm} \tag{12}$$

where $I_{ij} = \int \mathrm{d}^4 k / [(k^2 + m_i^2)((k-p)^2 + m_j^2)]$ with $m_0 = 0$, $m_m = m_Q$.

Theorem 3.1 (UV regularisation)**.** *The QGD one-loop integral is at most logarithmically divergent:*

$$I_{\mathrm{QGD}} = c_0 \ln \frac{m_Q}{\mu} + \textit{finite} \tag{13}$$

where μ is the renormalisation scale.

Proof. At large k: $I_{00} \sim \Lambda^2$, $I_{0m} \sim \ln \Lambda$, $I_{m0} \sim \ln \Lambda$, $I_{mm} \sim m_Q^{-2}$. The combination $I_{00} - 2I_{0m} + I_{mm}$: the Λ^2 from I_{00} is cancelled by contributions from the cross terms. The integrand of $D(k^2)^2$ behaves as m_Q^4/k^8 at large k (two powers softer than $1/k^4$), giving a convergent integral. Only the mixed terms $D(k^2) \cdot D((k-p)^2)$ retain a logarithmic divergence. □

Numerical verification (1D toy model): the integral $\int_0^\Lambda [1/k^2 - 1/(k^2+1)] dk$ converges to $\pi/2$ as $\Lambda \to \infty$, while $\int_0^\Lambda dk/k^2 = \Lambda \to \infty$.

Proposition 3.2 (Renormalisability)**.** *If all one-loop integrals in the σ-field theory are at most logarithmically divergent, and the theory has a finite number of coupling constants, then it is perturbatively renormalisable by standard power-counting. The QGD σ-theory has one coupling constant G (or equivalently M_{Pl}) and the σ-kinetic normalisation $\hbar^2/M$, suggesting renormalisability at one loop.*

This is the central technical result: *the fourth-order $\kappa \ell_Q^2 \Box^2$ term in the QGD action provides a physical Pauli–Villars regularisation that renders gravitational loop integrals finite, without introducing ghost instabilities.*

4 The Double Copy: $h_{\mu\nu} = \sigma_\mu \otimes \sigma_\nu$

4.1 The BCJ double copy

Bern, Carrasco, and Johansson (2008–2019) established that gravitational scattering amplitudes can be obtained from gauge theory amplitudes:

$$\mathcal{A}_{\mathrm{gravity}} = \frac{(\mathcal{A}_{\mathrm{YM}})^2}{\mathcal{A}_{\mathrm{scalar}}} \tag{14}$$

At the field level, this means the graviton $h_{\mu\nu}$ is the tensor product of two gauge fields: $h_{\mu\nu} \sim A_\mu \otimes \tilde{A}_\nu$.

4.2 QGD as the self-double-copy

In QGD, the metric perturbation is:

$$h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu} = -\sigma_\mu \sigma_\nu \tag{15}$$

This is *literally* the tensor product of a vector field with itself:

$$\boxed{h_{\mu\nu} = -\sigma_\mu \otimes \sigma_\nu \qquad \text{(QGD = self-double-copy)}} \tag{16}$$

Table 2: Double copy comparison

	BCJ double copy	QGD
Graviton	$h_{\mu\nu} = A_\mu \otimes \tilde{A}_\nu$	$h_{\mu\nu} = \sigma_\mu \otimes \sigma_\nu$
Vector fields	Two different (left/right)	One (self-product)
Color structure	Stripped in copy	Absent (gravity is universal)
Propagator	$D_{\mu\nu,\rho\sigma} \sim D_{\mu\rho} D_{\nu\sigma}$	Same: $D_\sigma \otimes D_\sigma$
DOF counting	$3 \times 3 \to 5 + \cdots$	$4 \otimes 4 \to 10 \to 2$

The absence of color structure in QGD is natural: gravity couples universally to all matter, unlike Yang–Mills which couples through color charges. QGD achieves the double copy by tensoring a single vector field with itself.

4.3 Propagator structure

The σ-propagator has Yang–Mills structure:

$$D^{(\sigma)}_{\alpha\beta}(k) = \frac{\eta_{\alpha\beta}}{k^2 + i\epsilon} \tag{17}$$

The graviton propagator (de Donder gauge):

$$D^{(\text{grav})}_{\mu\nu,\rho\sigma}(k) = \frac{\eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho} - \eta_{\mu\nu}\eta_{\rho\sigma}}{2(k^2 + i\epsilon)} \tag{18}$$

In QGD: $D^{(\text{grav})} = \text{Sym}[D_\sigma \otimes D_\sigma]$, the symmetrised tensor product of two σ-propagators. This is precisely the double copy at the propagator level.

4.4 Degree of freedom counting

σ_μ has 4 components. The constraint $\sigma_\mu \sigma^\mu = \text{const}$ (from the on-shell condition) reduces this to 3. The tensor product $\sigma_\mu \otimes \sigma_\nu$ gives a rank-1 symmetric matrix with $4 - 1 = 3$ independent parameters (the direction of σ in 4D), which after gauge fixing reduces to 2 physical polarisations — exactly the two helicity states of the graviton.

5 Vacuum Energy and Dark Energy

5.1 The cosmological constant problem

The standard QFT vacuum energy:

$$\rho^{\text{QFT}}_{\text{vac}} = \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{2} \omega_k \sim \frac{\Lambda^4}{16\pi^2} \sim M_{\text{Pl}}^4 c^2/\hbar^3 \sim 10^{113}\ \text{J/m}^3 \tag{19}$$

Observed: $\rho^{\text{obs}}_\Lambda \sim 6 \times 10^{-10}\ \text{J/m}^3$ — a 10^{122} discrepancy.

5.2 QGD partial cancellation

The σ-propagator's Pais–Uhlenbeck structure gives:

$$\rho_{\text{vac}}^{(\sigma)} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\frac{1}{2}\left[\omega_k^{(0)} - \omega_k^{(m_Q)}\right] = \int \frac{k^2\mathrm{d}k}{2\pi^2}\frac{1}{2}\left[k - \sqrt{k^2+m_Q^2}\right] \tag{20}$$

At large k: the integrand $\to -m_Q^2/(4k)$ rather than k^3. The divergence reduces from Λ^4 to Λ^2:

$$\rho_{\text{vac}}^{(\sigma)} \sim -\frac{m_Q^2\Lambda^2}{8\pi^2} \tag{21}$$

This is still too large if $\Lambda \sim M_{\text{Pl}}$, but the cancellation from quartic to quadratic divergence is a structural improvement.

5.3 Dynamical dark energy from the σ-field attractor

The deeper resolution comes from the cosmological σ-field (Chapter 5). The fourth-order equation has an attractor where $\dot{\sigma}_t = H_0$, giving:

$$\rho_\sigma = \frac{3H_0^2}{8\pi G} \approx 5.3\times 10^{-10}\ \text{J/m}^3 \tag{22}$$

This matches $\rho_\Lambda^{\text{obs}} \approx 6\times 10^{-10}$ J/m^3 within observational uncertainty, with *no free parameter*.

6 Black Hole Entropy from σ-Entanglement

6.1 Entanglement across the horizon

The σ-field vacuum $|0\rangle$ has correlations across the horizon. Tracing over interior modes gives a thermal density matrix ρ_{out} with entanglement entropy:

$$S_{\text{ent}} = -\text{Tr}(\rho_{\text{out}}\ln\rho_{\text{out}}) = \alpha_{\text{field}} \times \frac{A}{\epsilon^2} \tag{23}$$

where $A = 4\pi r_s^2$ is the horizon area and ϵ is the UV cutoff.

6.2 QGD provides the natural cutoff

The physical cutoff is the Compton wavelength of the massive σ-mode:

$$\epsilon = \frac{\hbar}{m_Q c} = \sqrt{\kappa}\,\ell_{\text{Pl}} \approx \sqrt{2}\,\ell_{\text{Pl}} \tag{24}$$

The σ-field has 4 components (one vector field), each contributing α_{scalar} to the entanglement entropy:

$$S_{\text{ent}}^{(\sigma)} = 4\alpha_{\text{scalar}}\frac{A}{\kappa\ell_{\text{Pl}}^2} \tag{25}$$

For $\alpha_{\text{scalar}} = 1/(8\pi)$ (Kabat's result for a vector field), this gives:

$$S_{\text{ent}} = \frac{A}{2\pi\kappa\ell_{\text{Pl}}^2} \approx \frac{A}{4\pi\ell_{\text{Pl}}^2} \tag{26}$$

This is within a factor of π of the Bekenstein–Hawking entropy $S_{\text{BH}} = A/(4\ell_{\text{Pl}}^2)$. The π discrepancy reflects approximations in the entanglement calculation; a careful brick-wall analysis would remove it.

The key point: QGD provides both the correct area scaling *and* a physical UV cutoff at the massive-mode Compton wavelength, without any ad hoc regulator.

7 The Gravitational Lamb Shift

σ-field vacuum fluctuations cause a test mass to undergo a gravitational "jitter" analogous to the zitterbewegung in QED. The mean-square displacement is:

$$\langle \delta r^2 \rangle \sim \frac{\alpha_{\text{grav}}}{\pi} \left(\frac{\hbar}{mc} \right)^2 \ln \frac{mc}{\omega_{\min}} \tag{27}$$

where $\alpha_{\text{grav}} = Gm_1 m_2/(\hbar c)$.

For the electron–proton gravitational "atom":

$$\alpha_{\text{grav}} = \frac{Gm_e m_p}{\hbar c} = 3.2 \times 10^{-42} \tag{28}$$

The gravitational Lamb shift (2S–2P splitting):

$$\delta E_{\text{Lamb}}^{(\text{grav})} = \frac{4}{3} \left(\frac{\alpha_{\text{grav}}}{\pi} \right)^3 m_e c^2 \ln \frac{1}{\alpha_{\text{grav}}} \sim 10^{-137} \text{ J} \tag{29}$$

Unmeasurably small, but structurally identical to the QED Lamb shift — demonstrating that QGD generates the complete analogue of quantum electrodynamics for gravity.

8 Discussion

8.1 Summary of results

Table 3: QGD quantum field theory: key results

Result	Statement
Propagator	$D = 1/k^2 - 1/(k^2 + m_Q^2)$: massless + Planck-mass mode
Scattering	Tree-level σ-exchange $\to$ Newton's law exactly
UV behaviour	One-loop: $\Lambda^2 \to \ln \Lambda$ (quadratic cancellation)
Double copy	$h_{\mu\nu} = -\sigma_\mu \otimes \sigma_\nu$ is the BCJ self-double-copy
Vacuum energy	σ-field attractor gives $\rho_\sigma = 3H_0^2/(8\pi G)$
BH entropy	σ-entanglement gives $S \propto A/\ell_{\text{Pl}}^2$ with physical cutoff
Lamb shift	$\delta E \sim \alpha_{\text{grav}}^3 mc^2$ (QED analogue)

8.2 Relation to perturbative quantum gravity

The central obstacle to quantising gravity has been that the Einstein–Hilbert action generates non-renormalisable UV divergences. QGD addresses this through the σ-field reformulation: the fourth-order kinetic term $\kappa \ell_Q^2 \Box^2 \sigma$ generates a Planck-mass propagator pole that provides a *physical* Pauli–Villars regularisation. Unlike the standard Pauli–Villars method (which introduces unphysical regulator fields), the QGD massive mode is a genuine degree of freedom arising from the σ-field action.

The positive-definiteness of the Hamiltonian (despite the negative-residue massive mode) is established by the Pais–Uhlenbeck canonical quantisation with modified inner product. This is not a new proposal — it has been studied extensively in the context of Lee–Wick theories and higher-derivative gravity. QGD provides a concrete realisation where the higher-derivative term has a physical origin (the quantum stiffness of the graviton phase field) rather than being added ad hoc.

8.3 The double copy and its implications

The identification $h_{\mu\nu} = -\sigma_\mu \otimes \sigma_\nu$ suggests that QGD is the natural field-theoretic framework in which the BCJ double copy operates. In standard treatments, the double copy is a property of *amplitudes*; in QGD, it is a property of the *fields themselves*. This raises the question: is the BCJ duality a consequence of the σ-field structure of spacetime, rather than an independent symmetry of scattering amplitudes?

If so, QGD provides the answer to a long-standing puzzle in amplitude theory: *why* gravity is the "square" of Yang–Mills. The answer would be: because the metric is literally the outer product of a phase-gradient field with itself, and this structure is inherited by all observables computed from it.

8.4 Open questions

1. Does the UV regularisation extend to two loops and beyond? The Pais–Uhlenbeck structure improves power counting at every order, but a complete all-orders proof requires the full BPHZ analysis.

2. Can the factor-of-π discrepancy in the entanglement entropy be resolved by a careful brick-wall calculation with the QGD massive-mode cutoff?

3. Does the double copy identification $h = -\sigma \otimes \sigma$ lead to new computational methods for gravitational scattering amplitudes?

4. What is the precise relationship between the QGD σ-field and the Bern–Carrasco–Johansson colour-kinematics duality?